

SOLIDWORKS API (NET VBA C#) — Reference (154/337): IMate2-IMateControllerFeatureData

SOLIDWORKS API Documentation (converted from CHM)

IMate2 (Type)

SOLIDWORKS API Help

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IMate2 Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMate2 Interface

Allows access to various assembly mate parameters.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IMate2
```

Visual Basic (Usage)



```
Dim instance As IMate2
```

C#

```
public interface IMate2
```

C++/CLI

```
public interface class IMate2
```

Visual Basic for Applications (VBA) Syntax

See [Mate2](#).

Example

[Get Mates and Mate Entities \(C#\)](#)
[Get Mates and Mate Entities \(VB.NET\)](#)
[Get Mates and Mate Entities \(VBA\)](#)

Remarks

To get the IMate2 object, you must first get it as a [IFeature](#) object and then use [IFeature::GetSpecificFeature2](#) or QueryInterface to return the IMate2 interface.

The IMate2 object is actually a subfeature beneath a MateGroup feature. To access the IMate2 subfeature by traversing the FeatureManager design tree, you can call [IFeature::GetFirstSubFeature](#) or [IFeature::!GetFirstSubFeature](#) and then [IFeature::GetNextSubFeature](#) or [IFeature::!GetNextSubFeature](#) when the current feature type is MateGroup.

Accessors

[IAssemblyDoc::AddConcentricMateWithTolerance](#)
[IAssemblyDoc::AddDistanceMate](#)
[IAssemblyDoc::AddMate3](#)
[IAssemblyDoc::AutoMateRepair](#)
[IAssemblyDoc::CollectAllSupportiveMates](#)
[IComponent2::GetMates](#)
[IFeature::GetSpecificFeature2](#)
[IMate2::GetLinkedMate](#)
[IMateControllerFeatureData::Mates](#)
[IMateLoadReference::Mate](#)
[IMoveCopyBodyFeatureData::AddMate](#) and [IMoveCopyBodyFeatureData::!AddMate](#)

Access Diagram

[Mate2](#)

See Also

[IMate2 Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMateEntity2 Interface](#)
[IMateLoadReference Interface](#)
[IAssemblyDoc Interface](#)

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IMate2 Interface Members

[See Also](#) [Properties](#) [Methods](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IMate2 Interface

The following tables list the members exposed by [IMate2](#).

Public Properties

Name	Description
 Alignment	Gets the type of alignment for this mate.
 CanBeFlipped	Gets whether this distance or angle mate can be flipped.
 DisplayDimension	Obsolete. Superseded by IMate2.DisplayDimension2.
 DisplayDimension2	Gets the specified display dimension for this mate.
 DistanceFirstArcCondition	Gets the the first arc condition of this distance mate between cylindrical components.
 DistanceSecondArcCondition	Gets the the second arc condition of this distance mate between cylindrical components.
 Flipped	Gets or sets whether to flip the distance or angle mate.
 HasLoadBearingFaces	Gets whether this mate has load bearing faces.
 HasTreatInterferenceAsShrinkFit	Gets whether interference in this mate is treated as shrink/press fit.
 IsLoadBearingFacesBonded	Get whether the load bearing faces of this mate are bonded.
 LockMagneticMate	Gets or sets whether to lock this magnetic mate.
 MateLoadReference	Gets the mate load reference associated with this mate.
 MaximumVariation	Gets the maximum variation, in meters or radians, for the dimension of this distance or angle mate.
 MinimumVariation	Gets the minimum variation, in meters or radians, for the dimension of this distance or angle mate.
 Type	Gets the type of mate.

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Public Methods

Name	Description
 ForceMisalignment	Forces a misaligned mate condition for this concentric mate.
 GetConcentricAlignmentType	Gets the alignment type of this mate.
 GetCurrentMisalignmentDeviation	Gets the current misalignment deviation for the misaligned concentric mate.
 GetForce	Gets the magnitude and direction of the force applied to this mate.
 GetLinkedMate	Gets the linked mate of this concentric mate.
 GetMateEntityCount	Gets the number of entities for this mate.
 GetMaximumMisalignmentDeviation	Gets the maximum allowed misalignment deviation for this misaligned concentric mate.
 GetSupplementalFaces	Gets the faces in this mate.
 GetTorque	Gets the angle and the axis of the torque applied to this mate.
 GetUseMisalignedDeviationDocumentProperty	Gets whether to use the document property value for the maximum misalignment deviation of the misaligned concentric mate.

 IGetSupplementalFaces	Gets the faces in this mate.
 MateEntity	Gets an entity associated with a mate.
 RemoveMisalignment	Removes the misaligned mate condition of this concentric mate.
 SetConcentricAlignmentType	Sets the alignment type of this mate.
 SetForce	Sets the magnitude and direction of the force to apply to this mate.
 SetMaximumMisalignmentDeviation	Sets the maximum allowed misalignment deviation for this misaligned concentric mate.
 SetTorque	Sets the angle and the axis of the torque to apply to this mate.
 SetUseMisalignedDeviationDocumentProperty	Sets whether to use the document property value for the maximum misalignment deviation for the misaligned concentric mate.

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See Also

[IMate2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMateEntity2 Interface](#)
[IMateLoadReference Interface](#)
[IAssemblyDoc Interface](#)

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IMate2 Interface Methods

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : [IMate2 Interface](#)

For a list of all members of this type, see [IMate2 members](#).

Public Methods

Name	Description
 ForceMisalignment	Forces a misaligned mate condition for this concentric mate.
 GetConcentricAlignmentType	Gets the alignment type of this mate.
 GetCurrentMisalignmentDeviation	Gets the current misalignment deviation for the misaligned concentric mate.
 GetForce	Gets the magnitude and direction of the force applied to this mate.
 GetLinkedMate	Gets the linked mate of this concentric mate.
 GetMateEntityCount	Gets the number of entities for this mate.
 GetMaximumMisalignedDeviation	Gets the maximum allowed misalignment deviation for this misaligned concentric mate.
 GetSupplementalFaces	Gets the faces in this mate.
 GetTorque	Gets the angle and the axis of the torque applied to this mate.
 GetUseMisalignedDeviationDocumentProperty	Gets whether to use the document property value for the maximum misalignment deviation of the misaligned concentric mate.
 IGetSupplementalFaces	Gets the faces in this mate.
 MateEntity	Gets an entity associated with a mate.
 RemoveMisalignment	Removes the misaligned mate condition of this concentric mate.
 SetConcentricAlignmentType	Sets the alignment type of this mate.
 SetForce	Sets the magnitude and direction of the force to apply to this mate.
 SetMaximumMisalignedDeviation	Sets the maximum allowed misalignment deviation for this misaligned concentric mate.
 SetTorque	Sets the angle and the axis of the torque to apply to this mate.
 SetUseMisalignedDeviationDocumentProperty	Sets whether to use the document property value for the maximum misalignment deviation for the misaligned concentric mate.

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See Also

[IMate2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMateEntity2 Interface](#)

IMate2 Interface Properties

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IMate2 Interface

For a list of all members of this type, see [IMate2 members](#).

Public Properties

Name	Description
 Alignment	Gets the type of alignment for this mate.
 CanBeFlipped	Gets whether this distance or angle mate can be flipped.
 DisplayDimension	Obsolete. Superseded by IMate2.DisplayDimension2
 DisplayDimension2	Gets the specified display dimension for this mate.
 DistanceFirstArcCondition	Gets the the first arc condition of this distance mate between cylindrical components.
 DistanceSecondArcCondition	Gets the the second arc condition of this distance mate between cylindrical components.
 Flipped	Gets or sets whether to flip the distance or angle mate.
 HasLoadBearingFaces	Gets whether this mate has load bearing faces.
 HasTreatInterferenceAsShrinkFit	Gets whether interference in this mate is treated as shrink/press fit.
 IsLoadBearingFacesBonded	Get whether the load bearing faces of this mate are bonded.
 LockMagneticMate	Gets or sets whether to lock this magnetic mate.
 MateLoadReference	Gets the mate load reference associated with this mate.
 MaximumVariation	Gets the maximum variation, in meters or radians, for the dimension of this distance or angle mate.
 MinimumVariation	Gets the minimum variation, in meters or radians, for the dimension of this distance or angle mate.
 Type	Gets the type of mate.

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See Also

[IMate2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMateEntity2 Interface](#)
[IMateLoadReference Interface](#)
[IAssemblyDoc Interface](#)

Alignment Property (IMate2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : Alignment Property (IMate2)

Gets the type of alignment for this mate.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Alignment As System.Integer

Visual Basic (Usage)

```
Dim instance As IMate2  
Dim value As System.Integer  
  
value = instance.Alignment
```

C#



```
System.int Alignment {get;}
```

C++/CLI

```
property System.int Alignment {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of alignment as defined in [swMateAlign_e](#)

Visual Basic for Applications (VBA) Syntax

See [Mate2::Alignment](#).

Example

[Edit Mate \(VBA\)](#)
[Get Mate Definition \(VBA\)](#)
[Get Mates and Mate Entities \(C#\)](#)
[Get Mates and Mate Entities \(VB.NET\)](#)
[Get Mates and Mate Entities \(VBA\)](#)

See Also

[IMate2 Interface](#)
[IMate2 Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

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CanBeFlipped Property (IMate2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : CanBeFlipped Property (IMate2)

Gets whether this distance or angle mate can be flipped.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property CanBeFlipped As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As System.Boolean  
  
value = instance.CanBeFlipped
```

C#

```
System.bool CanBeFlipped {get;}
```

C++/CLI

```
property System.bool CanBeFlipped {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if mate can be flipped, false if not

Visual Basic for Applications (VBA) Syntax

See [Mate2::CanBeFlipped](#).

Example

[Edit Mate \(VBA\)](#)
[Get Mate Definition \(VBA\)](#)
[Get Mates and Mate Entities \(C#\)](#)
[Get Mates and Mate Entities \(VB.NET\)](#)
[Get Mates and Mate Entities \(VBA\)](#)

Remarks

Use [IMate2::Type](#) to determine the type of mate.

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::Flipped Property](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

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DisplayDimension Property (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : DisplayDimension Property (IMate2)

Obsolete. Superseded by [IMate2::DisplayDimension2](#).

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property DisplayDimension As DisplayDimension
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As DisplayDimension  
  
value = instance.DisplayDimension
```

C#

```
DisplayDimension DisplayDimension {get;}
```

C++/CLI

```
property DisplayDimension^ DisplayDimension {  
    DisplayDimension^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Mate2::DisplayDimension](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)

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DisplayDimension2 Property (IMate2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : DisplayDimension2 Property (IMate2)

Gets the specified display dimension for this mate.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property DisplayDimension2( _  
    ByVal Index As System.Integer _  
) As DisplayDimension
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim Index As System.Integer  
Dim value As DisplayDimension  
  
value = instance.DisplayDimension2(Index)
```

C#

```
DisplayDimension DisplayDimension2(  
    System.int Index  
) {get;}
```

C++/CLI

```
property DisplayDimension^ DisplayDimension2 {  
    DisplayDimension^ get(System.int Index);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Number indicating which mate's display dimension to get (see **Remarks**)

Property Value

[Display dimension](#) for the specified mate

Visual Basic for Applications (VBA) Syntax

See [Mate2::DisplayDimension2](#).

Example

[Change Dimensions of Gear Mate \(VBA\)](#)
[Edit Mate \(VBA\)](#)

Remarks

This property returns the first display dimension for all types of mates and the second display dimension for gear mates only.

A gear mate has two feature dimensions whose values form the gear ratio. All other mates return NULL/Nothing for the second display dimension.

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::MaximumVariation Property \(\)](#)
[IMate2::MinimumVariation Property \(\)](#)

Availability

DistanceFirstArcCondition Property (IMate2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : DistanceFirstArcCondition Property (IMate2)

Gets the first arc condition of this distance mate between cylindrical components.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property DistanceFirstArcCondition As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IMate2
Dim value As System.Integer

value = instance.DistanceFirstArcCondition
```

C#

```
System.int DistanceFirstArcCondition {get;}
```

C++/CLI

```
property System.int DistanceFirstArcCondition {
    System.int get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Arc condition as defined by [swDistanceMateArcConditions](#) e

Visual Basic for Applications (VBA) Syntax

See [Mate2::DistanceFirstArcCondition](#).

Example

[Add and Edit Distance Mate \(VBA\)](#)
[Add and Edit Distance Mate \(VB.NET\)](#)
[Add and Edit Distance Mate \(C#\)](#)

Remarks

This property is valid only for distance mates between:

- two cylindrical faces
- or -
- one cylindrical face and one axis.

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::DistanceSecondArcCondition Property \(\)](#)
[IAssemblyDoc::AddDistanceMate Method \(\)](#)
[IAssemblyDoc::EditDistanceMate Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

DistanceSecondArcCondition Property (IMate2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : DistanceSecondArcCondition Property (IMate2)

Gets the the second arc condition of this distance mate between cylindrical components.

Visual Basic Syntax

Visual Basic (Declaration)

```
ReadOnly Property DistanceSecondArcCondition As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As System.Integer
```

```
value = instance.DistanceSecondArcCondition
```

C#

```
System.int DistanceSecondArcCondition {get;}
```

C++/CLI

```
property System.int DistanceSecondArcCondition {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Arc condition as defined by [swDistanceMateArcConditions](#) e

Visual Basic for Applications (VBA) Syntax

See [Mate2::DistanceSecondArcCondition](#).

Example

[Add and Edit Distance Mate \(VBA\)](#)
[Add and Edit Distance Mate \(VB.NET\)](#)
[Add and Edit Distance Mate \(C#\)](#)

Remarks

This property is valid only for distance mates between:

- two cylindrical faces
- OR -
- one cylindrical face and one axis.

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::DistanceFirstArcCondition Property \(\)](#)
[IAssemblyDoc::AddDistanceMate Method \(\)](#)
[IAssemblyDoc::EditDistanceMate Method \(\)](#)

Availability

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Flipped Property (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : Flipped Property (IMate2)

Gets or sets whether to flip the distance or angle mate.

.NET Syntax

Visual Basic (Declaration)

Property Flipped As System.Boolean

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As System.Boolean  
  
instance.Flipped = value  
value = instance.Flipped
```

C#

```
System.bool Flipped {get; set;}
```

C++/CLI

```
property System.bool Flipped {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to flip the mate, false to not

Visual Basic for Applications (VBA) Syntax

See [Mate2::Flipped](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::CanBeFlipped Property](#)
[IMate2::Type Property](#)

Availability

SOLIDWORKS 2004 SP4, Revision Number 12.4

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ForceMisalignment Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : ForceMisalignment Method (IMate2)

Forces a misaligned mate condition for this concentric mate.

.NET Syntax

Visual Basic (Declaration)

Function ForceMisalignment() As System.Boolean

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As System.Boolean  
  
value = instance.ForceMisalignment()
```

C#

```
System.bool ForceMisalignment()
```

C++/CLI

```
System.bool ForceMisalignment();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the misaligned mate condition is successfully created, false if not

Visual Basic for Applications (VBA) Syntax

See [Mate2::ForceMisalignment](#).

Example

```
'VBA

' Open an assembly with a misaligned concentric mate, Concentric1

Dim swfeature As SldWorks.Feature
Dim swmate As SldWorks.Mate2
Sub main()

    Set swApp = Application.SldWorks
    Set swasm = swApp.ActiveDoc
    Set Swfeat = swasm.FeatureByName("Concentric1")
    Set swmate = Swfeat.GetSpecificFeature2
    Debug.Print "Concentric mate type as defined in swConcentricAlignmentType_e: " &
    swmate.GetConcentricAlignmentType

    'Remove the misaligned mate condition
    swmate.RemoveMisalignment

    'Create the misaligned mate condition
    swmate.ForceMisalignment

End Sub
```

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::RemoveMisalignment Method \(\)](#)

Availability

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GetConcentricAlignmentType Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : GetConcentricAlignmentType Method (IMate2)

Gets the alignment type of this mate.

.NET Syntax

Visual Basic (Declaration)

```
Function GetConcentricAlignmentType() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IMate2
Dim value As System.Integer

value = instance.GetConcentricAlignmentType()
```

C#

```
System.int GetConcentricAlignmentType()
```

C++/CLI

```
System.int GetConcentricAlignmentType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Alignment type as defined in [swConcentricAlignmentType_e](#)

Visual Basic for Applications (VBA) Syntax

See [Mate2::GetConcentricAlignmentType](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::SetConcentricAlignmentType Method \(\)](#)
[IMate2::GetCurrentMisalignedDeviation Method \(\)](#)
[IMate2::GetMaximumMisalignedDeviation Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

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GetCurrentMisalignedDeviation Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : GetCurrentMisalignedDeviation Method (IMate2)

Gets the current misalignment deviation for the misaligned concentric mate.

.NET Syntax

Visual Basic (Declaration)

```
Function GetCurrentMisalignedDeviation() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMate2
Dim value As System.Double

value = instance.GetCurrentMisalignedDeviation()
```

C#

```
System.double GetCurrentMisalignedDeviation()
```

C++/CLI

```
System.double GetCurrentMisalignedDeviation();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Current misalignment deviation

Visual Basic for Applications (VBA) Syntax

See [IMate2::GetCurrentMisalignedDeviation](#).

Remarks

If the current deviation of misalignment is greater than [IMate2::GetMaximumMisalignedDeviation](#), then the mate fails to solve.

The current deviation and the maximum deviation are cumulative across both concentric mates in the Hole Set. For example, if [IMate2::GetConcentricAlignmentType](#) is set to [swConcentricAlignmentType_e.swConcentricAlignSymmetric](#), the deviation from both holes is summed before comparison with the maximum deviation.

See Also

[IMate2 Interface](#)
[IMate2 Members](#)

Availability

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GetForce Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : GetForce Method (IMate2)

Gets the magnitude and direction of the force applied to this mate.

.NET Syntax

Visual Basic (Declaration)

```
Function GetForce( _  
    ByRef Magnitude As System.Double _  
) As MathVector
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim Magnitude As System.Double  
Dim value As MathVector  
  
value = instance.GetForce(Magnitude)
```

C#

```
MathVector GetForce(  
    out System.double Magnitude  
)
```

C++/CLI

```
MathVector^ GetForce(  
& [Out] System.double Magnitude  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Magnitude
Magnitude

Return Value

[Direction](#)

Visual Basic for Applications (VBA) Syntax

See [Mate2::GetForce](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::SetForce Method](#)
[IMate2::GetTorque Method](#)
[IMate2::SetTorque Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

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GetLinkedMate Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : GetLinkedMate Method (IMate2)

Gets the linked mate of this concentric mate.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLinkedMate() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As System.Object  
  
value = instance.GetLinkedMate()
```

C#

```
System.object GetLinkedMate()
```

C++/CLI

```
System.Object^ GetLinkedMate();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Mate](#)

Visual Basic for Applications (VBA) Syntax

See [Mate2::GetLinkedMate](#).

Remarks

This mate and its linked mate are siblings in a Hole Set, which is defined as two concentric mates between the same two components.

See Also

[IMate2 Interface](#)
[IMate2 Members](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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GetMateEntityCount Method (IMate2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : GetMateEntityCount Method (IMate2)

Gets the number of entities for this mate.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMateEntityCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As System.Integer  
  
value = instance.GetMateEntityCount()
```

C#

```
System.int GetMateEntityCount()
```

C++/CLI

```
System.int GetMateEntityCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of entities for this mate

Visual Basic for Applications (VBA) Syntax

See [Mate2::GetMateEntityCount](#).

Example

[Edit Mate \(VBA\)](#)
[Get Mate Definition \(VBA\)](#)

Remarks

Call this method before calling [IMate2::MateEntity](#) to get the number of entities in the mate.

See Also

[IMate2 Interface](#)
[IMate2 Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

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GetMaximumMisalignedDeviation Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : GetMaximumMisalignedDeviation Method (IMate2)

Gets the maximum allowed misalignment deviation for this misaligned concentric mate.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaximumMisalignedDeviation() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMate2
Dim value As System.Double

value = instance.GetMaximumMisalignedDeviation()
```

C#

```
System.double GetMaximumMisalignedDeviation()
```

C++/CLI

```
System.double GetMaximumMisalignedDeviation();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Maximum allowed misalignment deviation

Visual Basic for Applications (VBA) Syntax

See [Mate2::GetMaximumMisalignedDeviation](#).

Remarks

If [IMate2::GetCurrentMisalignedDeviation](#) is greater than the value returned by this method, then the mate fails to solve.

The current deviation and the maximum deviation are cumulative across both concentric mates in the Hole Set. For example, if [IMate2::GetConcentricAlignmentType](#) is set to [swConcentricAlignmentType_e.swConcentricAlignSymmetric](#), the deviation from both holes is summed before comparison with the maximum deviation.

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::GetUseMisalignedDeviationDocumentProperty Method \(\)](#)
[IMate2::SetMaximumMisalignedDeviation Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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GetSupplementalFaces Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : GetSupplementalFaces Method (IMate2)

Gets the faces in this mate.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSupplementalFaces( _
    ByVal WhichOne As System.Integer _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMate2
Dim WhichOne As System.Integer
Dim value As System.Object
```

```
value = instance.GetSupplementalFaces(WhichOne)
```

C#

```
System.object GetSupplementalFaces(  
    System.int WhichOne  
)
```

C++/CLI

```
System.Object^ GetSupplementalFaces(  
& System.int WhichOne  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhichOne
Number of components in this mate

Return Value

Array of [faces](#) in this mate

Visual Basic for Applications (VBA) Syntax

See [Mate2::GetSupplementalFaces](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::GetSupplementalFaces Method](#)
[IMate2::HasLoadBearingFaces Property](#)
[IMate2::IsLoadBearingFacesBonded Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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GetTorque Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : GetTorque Method (IMate2)

Gets the angle and the axis of the torque applied to this mate.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTorque( _  
    ByRef Angle As System.Double _  
) As MathVector
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim Angle As System.Double  
Dim value As MathVector
```

```
value = instance.GetTorque(Angle)
```

C#

```
MathVector GetTorque(  
    out System.double Angle  
)
```

C++/CLI

```
MathVector^ GetTorque(  
& [Out] System.double Angle
```

)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle
Angle

Return Value

[Axis](#)

Visual Basic for Applications (VBA) Syntax

See [Mate2::GetTorque](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::SetTorque Method](#)
[IMate2::GetForce Method](#)
[IMate2::SetForce Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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GetUseMisalignedDeviationDocumentProperty Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : GetUseMisalignedDeviationDocumentProperty Method (IMate2)

Gets whether to use the document property value for the maximum misalignment deviation of the misaligned concentric mate.

.NET Syntax

Visual Basic (Declaration)

```
Function GetUseMisalignedDeviationDocumentProperty() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As System.Boolean  
  
value = instance.GetUseMisalignedDeviationDocumentProperty()
```

C#

```
System.bool GetUseMisalignedDeviationDocumentProperty()
```

C++/CLI

```
System.bool GetUseMisalignedDeviationDocumentProperty();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True to use the document property value for maximum misalignment deviation, false to not

Visual Basic for Applications (VBA) Syntax

See [Mate2::GetUseMisalignedDeviationDocumentProperty](#).

Remarks

If this method returns false, then the maximum deviation used is returned by [IMate2::GetMaximumMisalignedDeviation](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::SetUseMisalignedDeviationDocumentProperty Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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HasLoadBearingFaces Property (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : HasLoadBearingFaces Property (IMate2)

Gets whether this mate has load bearing faces.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property HasLoadBearingFaces As System.Boolean

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As System.Boolean  
  
value = instance.HasLoadBearingFaces
```

C#

```
System.bool HasLoadBearingFaces {get;}
```

C++/CLI

```
property System.bool HasLoadBearingFaces {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the mate has load bearing faces, false if not

Visual Basic for Applications (VBA) Syntax

See [Mate2::HasLoadBearingFaces](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::IsLoadBearingFacesBonded Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

HasTreatInterferenceAsShrinkFit Property (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : HasTreatInterferenceAsShrinkFit Property (IMate2)

Gets whether interference in this mate is treated as shrink/press fit.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property HasTreatInterferenceAsShrinkFit As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMate2
Dim value As System.Boolean

value = instance.HasTreatInterferenceAsShrinkFit
```

C#

```
System.bool HasTreatInterferenceAsShrinkFit {get;}
```

C++/CLI

```
property System.bool HasTreatInterferenceAsShrinkFit {
    System.bool get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if interference is treated as shrink/press fit, false if not

Visual Basic for Applications (VBA) Syntax

See [Mate2::HasTreatInterferenceAsShrinkFit](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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IGetSupplementalFaces Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : IGetSupplementalFaces Method (IMate2)

Gets the faces in this mate.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSupplementalFaces( _
    ByVal WhichOne As System.Integer, _
    ByVal FaceCount As System.Integer _
) As Face2
```

Visual Basic (Usage)



```
Dim instance As IMate2
Dim WhichOne As System.Integer
Dim FaceCount As System.Integer
Dim value As Face2

value = instance.IGetSupplementalFaces(WhichOne, FaceCount)
```

C#

```
Face2 IGetSupplementalFaces(
    System.int WhichOne,
```

```
        System.int FaceCount
    )
```

C++/CLI

```
Face2^ IGetSupplementalFaces(
&    System.int WhichOne,
&    System.int FaceCount
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhichOne

Number of components in this mate

FaceCount

Number of faces in this mate

Return Value

- in-process, unmanaged C++: Pointer to an array of [faces](#) in this mate
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

See Also

[IMate2 Interface](#)

[IMate2 Members](#)

[IMate2::GetSupplementalFaces Method](#)

[IMate2::HasLoadBearingFaces Property](#)

[IMate2::IsLoadBearingFacesBonded Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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IsLoadBearingFacesBonded Property (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : IsLoadBearingFacesBonded Property (IMate2)

Get whether the load bearing faces of this mate are bonded.

Visual Basic Syntax

Visual Basic (Declaration)

```
ReadOnly Property IsLoadBearingFacesBonded As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMate2
```

```
Dim value As System.Boolean
```

```
value = instance.IsLoadBearingFacesBonded
```

C#

```
System.bool IsLoadBearingFacesBonded {get;}
```

C++/CLI

```
property System.bool IsLoadBearingFacesBonded {
    System.bool get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the load-bearing faces of this mate are bonded, false if not

Visual Basic for Applications (VBA) Syntax

See [Mate2::IsLoadBearingFacesBonded](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::HasLoadBearingFaces Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

LockMagneticMate Property (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : LockMagneticMate Property (IMate2)

Gets or sets whether to lock this magnetic mate.

.NET Syntax

Visual Basic (Declaration)

Property LockMagneticMate As System.Boolean

Visual Basic (Usage)



```
Dim instance As IMate2
Dim value As System.Boolean

instance.LockMagneticMate = value

value = instance.LockMagneticMate
```

C#

```
System.bool LockMagneticMate {get; set;}
```

C++/CLI

```
property System.bool LockMagneticMate {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to lock, false to not

Visual Basic for Applications (VBA) Syntax

See [Mate2::LockMagneticMate](#).

Remarks

This property is valid only if [IMate2::Type](#) is set to [swMateType_e.swMateMAGNETIC](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)

Availability

MateEntity Method (IMate2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : MateEntity Method (IMate2)

Gets an entity associated with a mate.

.NET Syntax

Visual Basic (Declaration)

```
Function MateEntity( _  
    ByVal Index As System.Integer _  
) As MateEntity2
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim Index As System.Integer  
Dim value As MateEntity2  
  
value = instance.MateEntity(Index)
```

C#

```
MateEntity2 MateEntity(  
    System.int Index  
)
```

C++/CLI

```
MateEntity2^ MateEntity(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the entity associated with the mate

Return Value

[Mate entity](#)

Visual Basic for Applications (VBA) Syntax

See [Mate2::MateEntity](#).

Example

[Edit Mate \(VBA\)](#)
[Get Mates and Mate Entities \(C#\)](#)
[Get Mates and Mate Entities \(VB.NET\)](#)
[Get Mates and Mate Entities \(VBA\)](#)

Remarks

To get the number of entities for this mate, call [IMate2::GetMateEntityCount](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)

Availability

MateLoadReference Property (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : MateLoadReference Property (IMate2)

Gets the mate load reference associated with this mate.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property MateLoadReference As MateLoadReference
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As MateLoadReference  
  
value = instance.MateLoadReference
```

C#

```
MateLoadReference MateLoadReference {get;}
```

C++/CLI

```
property MateLoadReference^ MateLoadReference {  
    MateLoadReference^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Mate load reference](#) for this mate, or Nothing or null if none exists

Visual Basic for Applications (VBA) Syntax

See [Mate2::MateLoadReference](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

MaximumVariation Property (IMate2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : MaximumVariation Property (IMate2)

Gets the maximum variation, in meters or radians, for the dimension of this distance or angle mate.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property MaximumVariation As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As System.Double
```

```
value = instance.MaximumVariation
```

C#

```
System.double MaximumVariation {get;}
```

C++/CLI

```
property System.double MaximumVariation {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Maximum variation for the dimension of this mate

Visual Basic for Applications (VBA) Syntax

See [Mate2::MaximumVariation](#).

Example

[Edit Mate \(VBA\)](#)
[Get Mate Definition \(VBA\)](#)

Remarks

This property is valid only for [IMate2::Type](#):

- swMateType_e.swMateANGLE
- OR -
- swMateType_e.swMateDISTANCE

For distance and angle mates:

Maximum_variation = maximum_value - dimension_value

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::DisplayDimension2 Property](#)
[IMate2::MinimumVariation Property](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MinimumVariation Property (IMate2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : MinimumVariation Property (IMate2)

Gets the minimum variation, in meters or radians, for the dimension of this distance or angle mate.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property MinimumVariation As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As System.Double  
  
value = instance.MinimumVariation
```

C#

```
System.double MinimumVariation {get;}
```

C++/CLI

```
property System.double MinimumVariation {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Minimum variation for the dimension of this mate

Visual Basic for Applications (VBA) Syntax

See [Mate2::MinimumVariation](#).

Example

[Edit Mate \(VBA\)](#)
[Get Mate Definition \(VBA\)](#)

Remarks

This property is valid only for [IMate2::Type](#):

- swMateType_e.swMateANGLE
- OR -
- swMateType_e.swMateDISTANCE

For distance and angle mates:

Minimum_variation = minimum_value - dimension_value

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::DisplayDimension2 Property](#)
[IMate2::MaximumVariation Property](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

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RemoveMisalignment Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : RemoveMisalignment Method (IMate2)

Removes the misaligned mate condition of this concentric mate.

.NET Syntax

Visual Basic (Declaration)

```
Function RemoveMisalignment() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As System.Boolean
```

```
value = instance.RemoveMisalignment()
```

C#

```
System.bool RemoveMisalignment()
```

```
System.bool RemoveMisalignment();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the misaligned mate condition is successfully removed, false if not

Visual Basic for Applications (VBA) Syntax

See [Mate2::RemoveMisalignment](#).

Example

```
'VBA

'Open an assembly with a misaligned concentric mate, Concentric1

Dim swfeature As SldWorks.Feature
Dim swmate As SldWorks.Mate2
Sub main()

    Set swApp = Application.SldWorks
    Set swasm = swApp.ActiveDoc
    Set Swfeat = swasm.FeatureByName("Concentric1")
    Set swmate = Swfeat.GetSpecificFeature2
    Debug.Print "Concentric mate type as defined in swConcentricAlignmentType_e: " &
    swmate.GetConcentricAlignmentType

    'Remove the misaligned mate condition
    swmate.RemoveMisalignment

    'Create the misaligned mate condition
    swmate.ForceMisalignment

End Sub
```

Remarks

After calling this method, one of the concentric mates may be unsolvable.

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::ForceMisalignment Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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SetConcentricAlignmentType Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : SetConcentricAlignmentType Method (IMate2)

Sets the alignment type of this mate.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetConcentricAlignmentType( _
    ByVal PositionType As System.Integer _
```

)

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim PositionType As System.Integer  
  
instance.SetConcentricAlignmentType(PositionType)
```

C#

```
void SetConcentricAlignmentType(  
    System.int PositionType  
)
```

C++/CLI

```
void SetConcentricAlignmentType(  
& System.int PositionType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PositionType
Alignment type as defined in [swConcentricAlignmentType_e](#)

Visual Basic for Applications (VBA) Syntax

See [Mate2::SetConcentricAlignmentType](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::GetConcentricAlignmentType Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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SetForce Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : SetForce Method (IMate2)

Sets the magnitude and direction of the force to apply to this mate.

.NET Syntax

Visual Basic (Declaration)

```
Function SetForce( _  
    ByVal Magnitude As System.Double, _  
    ByVal Direction As MathVector _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim Magnitude As System.Double  
Dim Direction As MathVector  
Dim value As System.Boolean  
  
value = instance.SetForce(Magnitude, Direction)
```

C#

```
System.bool SetForce(  
    System.double Magnitude,  
    MathVector Direction  
)
```

C++/CLI

```
System.bool SetForce(  
& System.double Magnitude,  
& MathVector^ Direction  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Magnitude
Magnitude
Direction
[Direction](#)

Return Value

True if the operation succeeds, false if not

Visual Basic for Applications (VBA) Syntax

See [Mate2::SetForce](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::GetForce Method](#)
[IMate2::GetTorque Method](#)
[IMate2::SetTorque Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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SetMaximumMisalignedDeviation Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : SetMaximumMisalignedDeviation Method (IMate2)

Sets the maximum allowed misalignment deviation for this misaligned concentric mate.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetMaximumMisalignedDeviation( _  
    ByVal MaximumDeviation As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim MaximumDeviation As System.Double  
  
instance.SetMaximumMisalignedDeviation(MaximumDeviation)
```

C#

```
void SetMaximumMisalignedDeviation(  
    System.double MaximumDeviation  
)
```

C++/CLI

```
void SetMaximumMisalignedDeviation(  
& System.double MaximumDeviation  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MaximumDeviation
Maximum allowed misalignment deviation

Visual Basic for Applications (VBA) Syntax

See [IMate2::SetMaximumMisalignedDeviation](#).

Remarks

This method is valid only if [IMate2::GetUseMisalignedDeviationDocumentProperty](#) returns false. Call [IMate2::SetUseMisalignedDeviationDocumentProperty](#) to set UseDocumentProperty to false if you want to use this method to set the maximum misaligned deviation for this mate.

If [IMate2::GetCurrentMisalignedDeviation](#) is greater than the value set by this method, then the mate fails to solve.

The current deviation and the maximum deviation are cumulative across both concentric mates in the Hole Set. For example, if [IMate2::GetConcentricAlignmentType](#) is set to [swConcentricAlignmentType_e.swConcentricAlignSymmetric](#), the deviation from both holes is summed before comparison with the maximum deviation.

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::GetMaximumMisalignedDeviation Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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SetTorque Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : SetTorque Method (IMate2)

Sets the angle and the axis of the torque to apply to this mate.

.NET Syntax

Visual Basic (Declaration)

```
Function SetTorque(  
    ByVal Angle As System.Double, _  
    ByVal Axis As MathVector _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim Angle As System.Double  
Dim Axis As MathVector  
Dim value As System.Boolean
```

```
value = instance.SetTorque(Angle, Axis)
```

C#

```
System.bool SetTorque(  
    System.double Angle,
```

```
MathVector Axis
)
```

C++/CLI

```
System.bool SetTorque(
& System.double Angle,
& MathVector^ Axis
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle
Angle
Axis
[Axis](#)

Return Value

True if the operation succeeds, false if not

Visual Basic for Applications (VBA) Syntax

See [Mate2::SetTorque](#).

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::GetTorque Method](#)
[IMate2::GetForce Method](#)
[IMate2::SetForce Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

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SetUseMisalignedDeviationDocumentProperty Method (IMate2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : SetUseMisalignedDeviationDocumentProperty Method (IMate2)

Sets whether to use the document property value for the maximum misalignment deviation for the misaligned concentric mate.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetUseMisalignedDeviationDocumentProperty( _
    ByVal UseDocumentProperty As System.Boolean _
)
```

Visual Basic (Usage)



```
Dim instance As IMate2
Dim UseDocumentProperty As System.Boolean

instance.SetUseMisalignedDeviationDocumentProperty(UseDocumentProperty)
```

C#

```
void SetUseMisalignedDeviationDocumentProperty(
    System.bool UseDocumentProperty
)
```

C++/CLI

```
void SetUseMisalignedDeviationDocumentProperty(  
& System.bool UseDocumentProperty  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UseDocumentProperty

True to use the document property value for maximum misalignment deviation, false to not

Visual Basic for Applications (VBA) Syntax

See [Mate2::SetUseMisalignedDeviationDocumentProperty](#).

Remarks

If [IMate2::GetCurrentMisalignedDeviation](#) is greater than the document property value of the maximum misalignment deviation, then the mate fails to solve.

The current deviation and the maximum deviation are cumulative across both concentric mates in the Hole Set. For example, if [IMate2::GetConcentricAlignmentType](#) is set to [swConcentricAlignmentType_e.swConcentricAlignSymmetric](#), the deviation from both holes is summed before comparison with the maximum deviation.

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::GetUseMisalignedDeviationDocumentProperty Method \(\)](#)
[IMate2::SetMaximumMisalignedDeviation Method \(\)](#)

Availability

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SOLIDWORKS API Help

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Type Property (IMate2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMate2 Interface](#) : Type Property (IMate2)

Gets the type of mate.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Type As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IMate2  
Dim value As System.Integer  
  
value = instance.Type
```

C#

```
System.int Type {get;}
```

C++/CLI

```
property System.int Type {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of mate as defined by [swMateType_e](#)

Visual Basic for Applications (VBA) Syntax

See [Mate2::Type](#).

Example

[Edit Mate \(VBA\)](#)
[Get Mate Definition \(VBA\)](#)
[Get Mates \(C#\)](#)
[Get Mates \(VB.NET\)](#)
[Get Mates \(VBA\)](#)
[Get Mates and Mate Entities \(C#\)](#)
[Get Mates and Mate Entities \(VB.NET\)](#)
[Get Mates and Mate Entities \(VBA\)](#)

See Also

[IMate2 Interface](#)
[IMate2 Members](#)
[IMate2::LockMagneticMate Property\(\)](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

IMateControllerFeatureData (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IMateControllerFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMateControllerFeatureData Interface

Allows access to a mate controller feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IMateControllerFeatureData
```

Visual Basic (Usage)



```
Dim instance As IMateControllerFeatureData
```

C#

```
public interface IMateControllerFeatureData
```

C++/CLI

```
public interface class IMateControllerFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [MateControllerFeatureData](#).

Example

'VBA

'=====

' Preconditions: Open an assembly document that contains a mechanical slot feature

' with a slot mate that has one of these constraints:

' - Distance Along Slot

' - Percent Along Slot

'

' Postconditions:

' 1. **Mate Controller (Position 3)** is added to the FeatureManager design tree.

' 2. Inspect the graphics area.

'

'=====

```
Dim swApp As SldWorks.SldWorks
```

```
Dim Part As SldWorks.ModelDoc2
```

```
Dim Assem As SldWorks.AssemblyDoc
```

```
Dim boolstatus As Boolean
```

```
Option Explicit
```

```
Sub main()
```

```
    Set swApp = Application.SldWorks
```

```
    Set Part = swApp.ActiveDoc
```

```
    Set Assem = Part
```

```
    Dim myModelView As Object
```

```
    Set myModelView = Part.ActiveView
```

```
    myModelView.FrameState = swWindowState_e.swWindowMaximized
```

```
    Part.ClearSelection2 True
```

```
    boolstatus = Assem.IsSupportedMatesAvailable
```

```
    Dim var As Variant
```

```
    var = Assem.CollectAllSupportiveMates
```

```
    Dim featMgr As SldWorks.FeatureManager
```

```

Set featMgr = Part.FeatureManager

Dim mateControllerObj As SldWorks.MateControllerFeatureData
Set mateControllerObj = featMgr.CreateDefinition(swFmMateController)

Call mateControllerObj.Initialize(var)

mateControllerObj.AddNewPosition ("Position1")
mateControllerObj.AddNewPosition ("Position2")
mateControllerObj.AddNewPosition ("Position3")

Dim position2ValArr(0) As Double
position2ValArr(0) = 0.0375

Call mateControllerObj.SetValues("Position2", position2ValArr)

Dim position3ValArr(0) As Double
position3ValArr(0) = 0.0625

Call mateControllerObj.SetValues("Position3", position3ValArr)

Dim mateContFeat As SldWorks.Feature
Set mateContFeat = featMgr.CreateFeature(mateControllerObj)

Dim mcObj2 As SldWorks.MateControllerFeatureData
Set mcObj2 = mateContFeat.GetDefinition()

Dim var1 As Variant
var1 = mcObj2.Mates

Dim position2ValArr2(0) As Double
position2ValArr2(0) = 0.015

Call mcObj2.SetValues("Position2", position2ValArr2)

Dim position3ValArr2(0) As Double
position3ValArr2(0) = 0.085

Call mcObj2.SetValues("Position3", position3ValArr2)

mateContFeat.ModifyDefinition mcObj2, Part, Nothing

End Sub

```

Example

[Create Mate Controller \(VB.NET\)](#)
[Create Mate Controller \(C#\)](#)

Remarks

To create a mate controller feature:

- Use `IAssemblyDoc::IsSupportedMatesAvailable` to determine whether supportive mates exist for the assembly.

If true:

1. Call [IAssemblyDoc::CollectAllSupportiveMates](#).
2. Pre-select the supportive mates to use to create the mate controller. Or call [IMateControllerFeatureData::Initialize](#), passing in the supportive mates. Mates passed into the Initialize method are given precedence over pre-selected mates. The array passed in must contain IDispatch objects.
3. Call [IFeatureManager::CreateDefinition](#) to create an IMateControllerFeatureData object.
4. Call [IMateControllerFeatureData::AddNewPosition](#) to add a new mate position to the mate controller.
5. Call [IMateControllerFeatureData::SetValues](#) to set the values for all mates in a given mate position. If values are not specified or invalid values are specified, then the mate controller feature is created with default mate values (i.e., values used when the mate was created).
6. Call [IFeatureManager::CreateFeature](#), passing in the IMateControllerFeatureData object.
7. Call [IFeatureManager::GetCreateFeatureErrors](#).

To edit a mate controller feature, call:

1. [IFeature::GetDefinition](#)
2. [IMateControllerFeatureData::AccessSelections](#)
3. [IFeature::ModifyDefinition](#)
4. [IMateControllerFeatureData::ReleaseSelectionAccess](#)

For more information about the Mate Controller feature, see the **SOLIDWORKS Design user-interface help > Assemblies > Mates > Mate Controller** topic.

Accessors

`IFeature::GetDefinition`

`IFeatureManager::CreateDefinition`

Access Diagram

[MateControllerFeatureData](#)

See Also

[IMateControllerFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IMateControllerFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMateControllerFeatureData Interface

For a list of all members of this type, see [IMateControllerFeatureData members](#).

Public Methods

Name	Description
 AccessSelections	Accesses the selections used to define this mate controller feature.
 AddConfigurationFromPosition	Adds a configuration from the specified mate position.
 AddNewPosition	Adds a mate position name to this mate controller.
 DeletePosition	Deletes the specified mate position from this mate controller.
 GetPositions	Gets all of the mate positions in this mate controller.
 GetValues	Gets the mate values for the specified mate position.
 Initialize	Initializes this mate controller with the specified mates.
 ReleaseSelectionAccess	Releases access to the selections for this loft feature.
 ReOrderPositions	Reorders the mate positions in this mate controller.
 SetValues	Sets the mate values for the specified mate position.

[Top](#)

See Also

[IMateControllerFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IMateControllerFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMateControllerFeatureData Interface

The following tables list the members exposed by [IMateControllerFeatureData](#).

Public Properties

Name	Description
 Mates	Gets or sets the mates in this mate controller.

[Top](#)

Public Methods

Name	Description
 AccessSelections	Accesses the selections used to define this mate controller feature.
 AddConfigurationFromPosition	Adds a configuration from the specified mate position.
 AddNewPosition	Adds a mate position name to this mate controller.
 DeletePosition	Deletes the specified mate position from this mate controller.
 GetPositions	Gets all of the mate positions in this mate controller.

		
	GetValues	Gets the mate values for the specified mate position.
	Initialize	Initializes this mate controller with the specified mates.
	ReleaseSelectionAccess	Releases access to the selections for this loft feature.
	ReOrderPositions	Reorders the mate positions in this mate controller.
	SetValues	Sets the mate values for the specified mate position.

[Top](#)

 See Also

[IMateControllerFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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IMateControllerFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMateControllerFeatureData Interface

For a list of all members of this type, see [IMateControllerFeatureData members](#).

 Public Properties

Name	Description
 Mates	Gets or sets the mates in this mate controller.

[Top](#)

 See Also

[IMateControllerFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

AccessSelections Method (IMateControllerFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMateControllerFeatureData Interface](#) : AccessSelections Method (IMateControllerFeatureData)

Accesses the selections used to define this mate controller feature.

 .NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections(  
    ByVal TopDoc As System.Object, _  
    ByVal Component As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMateControllerFeatureData  
Dim TopDoc As System.Object  
Dim Component As System.Object  
Dim value As System.Boolean  
  
value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(  
    System.object TopDoc,  
    System.object Component  
)
```

C++/CLI

```
System.bool AccessSelections(
& System.Object^ TopDoc,
& System.Object^ Component
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Top-level document (see **Remarks**)

Component

Component in which the feature is to be modified (see **Remarks**)

Return Value

True if the selections were accessed successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [MateControllerFeatureData::AccessSelections](#).

Remarks

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) if you modified the feature
- [IMateControllerFeatureData::ReleaseSelectionAccess](#) if you did not

See Also

[IMateControllerFeatureData Interface](#)
[IMateControllerFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddConfigurationFromPosition Method (IMateControllerFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMateControllerFeatureData Interface](#) : AddConfigurationFromPosition Method (IMateControllerFeatureData)

Adds a configuration from the specified mate position.

.NET Syntax

Visual Basic (Declaration)

```
Function AddConfigurationFromPosition( _
    ByVal PositionName As System.String, _
    ByVal IsUpdate As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMateControllerFeatureData
Dim PositionName As System.String
Dim IsUpdate As System.Boolean
Dim value As System.Boolean
```

```
value = instance.AddConfigurationFromPosition(PositionName, IsUpdate)
```

C#

```
System.bool AddConfigurationFromPosition(
    System.string PositionName,
    System.bool IsUpdate
)
```

C++/CLI

```
System.bool AddConfigurationFromPosition(
    & System.String^ PositionName,
    & System.bool IsUpdate
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PositionName
Name of mate position
IsUpdate
True if an update, false if not

Return Value

True if configuration successfully added, false if not

Visual Basic for Applications (VBA) Syntax

See [MateControllerFeatureData::AddConfigurationFromPosition](#).

See Also

[IMateControllerFeatureData Interface](#)
[IMateControllerFeatureData Members](#)

Availability

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SOLIDWORKS API Help

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AddNewPosition Method (IMateControllerFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMateControllerFeatureData Interface](#) : AddNewPosition Method (IMateControllerFeatureData)

Adds a mate position name to this mate controller.

.NET Syntax

Visual Basic (Declaration)

```
Function AddNewPosition( _
    ByVal PositionName As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMateControllerFeatureData
Dim PositionName As System.String
Dim value As System.Boolean

value = instance.AddNewPosition(PositionName)
```

C#

```
System.bool AddNewPosition(
    System.string PositionName
)
```

C++/CLI

```
System.bool AddNewPosition(  
& System.String^ PositionName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PositionName
Name of mate position to add

Return Value

True if mate position successfully created, false if not

Visual Basic for Applications (VBA) Syntax

See [MateControllerFeatureData::AddNewPosition](#).

Example

See the [IMateControllerFeatureData](#) examples.

See Also

[IMateControllerFeatureData Interface](#)
[IMateControllerFeatureData Members](#)

Availability

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DeletePosition Method (IMateControllerFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMateControllerFeatureData Interface](#) : DeletePosition Method (IMateControllerFeatureData)

Deletes the specified mate position from this mate controller.

.NET Syntax

Visual Basic (Declaration)

```
Function DeletePosition(  
    ByVal PositionName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMateControllerFeatureData  
Dim PositionName As System.String  
Dim value As System.Boolean  
  
value = instance.DeletePosition(PositionName)
```

C#

```
System.bool DeletePosition(  
    System.string PositionName  
)
```

C++/CLI

```
System.bool DeletePosition(  
& System.String^ PositionName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PositionName

Name of mate position to delete

Return Value

True if mate position successfully deleted, false if not

Visual Basic for Applications (VBA) Syntax

See [MateControllerFeatureData.DeletePosition](#).

Remarks

Before calling this method, use [IMateControllerFeatureData.GetPositions](#) to get the existing mate positions.

See Also

[IMateControllerFeatureData Interface](#)

[IMateControllerFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

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GetPositions Method (IMateControllerFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMateControllerFeatureData Interface](#) : GetPositions Method (IMateControllerFeatureData)

Gets all of the mate positions in this mate controller.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPositions() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMateControllerFeatureData  
Dim value As System.Object
```

```
value = instance.GetPositions()
```

C#

```
System.object GetPositions()
```

C++/CLI

```
System.Object^ GetPositions();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of mate positions

Visual Basic for Applications (VBA) Syntax

See [MateControllerFeatureData.GetPositions](#).

See Also

[IMateControllerFeatureData Interface](#)

[IMateControllerFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

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GetValues Method (IMateControllerFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMateControllerFeatureData Interface](#) : GetValues Method (IMateControllerFeatureData)

Gets the mate values for the specified mate position.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetValues( _  
    ByVal PositionName As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMateControllerFeatureData  
Dim PositionName As System.String  
Dim value As System.Object  
  
value = instance.GetValues(PositionName)
```

C#

```
System.object GetValues(  
    System.string PositionName  
)
```

C++/CLI

```
System.Object^ GetValues(  
& System.String^ PositionName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PositionName
Name of mate position

Return Value

Array of values for all mates in PositionName

Visual Basic for Applications (VBA) Syntax

See [MateControllerFeatureData::GetValues](#).

See Also

[IMateControllerFeatureData Interface](#)
[IMateControllerFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments on this topic.](#)

Initialize Method (IMateControllerFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMateControllerFeatureData Interface](#) : Initialize Method (IMateControllerFeatureData)

Initializes this mate controller with the specified mates.

Visual Basic Syntax

Visual Basic (Declaration)

```
Sub Initialize( _  
    ByVal Mates As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As IMateControllerFeatureData  
Dim Mates As System.Object  
  
instance.Initialize(Mates)
```

C#

```
void Initialize(  
    System.object Mates  
)
```

C++/CLI

```
void Initialize(  
& System.Object^ Mates  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Mates

Array of supportive [mates](#)

Visual Basic for Applications (VBA) Syntax

See [MateControllerFeatureData::Initialize](#).

Example

See the [IMateControllerFeatureData](#) examples.

Remarks

Before calling this method, use [IAssemblyDoc::IsSupportedMatesAvailable](#) and [IAssemblyDoc::CollectAllSupportiveMates](#) to determine appropriate mates with which to initialize this mate controller.

Mates passed to this method are given precedence over pre-selected mates. If this method is not called or invalid mates are passed to it, then pre-selected mates are used.

For .NET applications, you must marshal the Mates array to IDispatch object arrays before calling this method. See [IDispatch Object Arrays as Input in .NET](#).

See Also

[IMateControllerFeatureData Interface](#)
[IMateControllerFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

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Mates Property (IMateControllerFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMateControllerFeatureData Interface](#) : Mates Property (IMateControllerFeatureData)

Gets or sets the mates in this mate controller.

Visual Basic Syntax

Visual Basic (Declaration)

Property Mates As System.Object

Visual Basic (Usage)



```
Dim instance As IMateControllerFeatureData
Dim value As System.Object

instance.Mates = value
value = instance.Mates
```

C#

```
System.Object Mates {get; set;}
```

C++/CLI

```
property System.Object^ Mates {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of supportive [mates](#)

Visual Basic for Applications (VBA) Syntax

See [MateControllerFeatureData::Mates](#).

Example

See the [IMateControllerFeatureData](#) examples.

Remarks

This property is only valid after the mate controller is created.

To create a mate controller see the Remarks section of IMateControllerFeatureData.

See Also

[IMateControllerFeatureData Interface](#)
[IMateControllerFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReleaseSelectionAccess Method (IMateControllerFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMateControllerFeatureData Interface](#) : ReleaseSelectionAccess Method (IMateControllerFeatureData)

Releases access to the selections for this loft feature.

Visual Basic Syntax

Visual Basic (Declaration)

Sub ReleaseSelectionAccess()

```
Dim instance As IMateControllerFeatureData
instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [MateControllerFeatureData::ReleaseSelectionAccess](#).

Remarks

[IMateControllerFeatureData::AccessSelections](#) puts the model into a rollback state to allow access to the selections that define this feature. Use this method to restore the rollback state if you did not modify the feature or use [IFeature::ModifyDefinition](#) if you did.

See Also

[IMateControllerFeatureData Interface](#)
[IMateControllerFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReOrderPositions Method (IMateControllerFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMateControllerFeatureData Interface](#) : ReOrderPositions Method (IMateControllerFeatureData)

Reorders the mate positions in this mate controller.

.NET Syntax

Visual Basic (Declaration)

```
Sub ReOrderPositions( _
    ByVal Positions As System.Object _
)
```

Visual Basic (Usage)

Copy Code

```
Dim instance As IMateControllerFeatureData
Dim Positions As System.Object

instance.ReOrderPositions(Positions)
```

C#

```
void ReOrderPositions(
    System.object Positions
)
```

C++/CLI

```
void ReOrderPositions(
    & System.Object^ Positions
```

)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Positions

Array of re-ordered mate positions

Visual Basic for Applications (VBA) Syntax

See [MateControllerFeatureData::ReOrderPositions](#).

Remarks

Before calling this method, use [IMateControllerFeatureData::GetPositions](#) to get the existing mate positions.

See Also

[IMateControllerFeatureData Interface](#)

[IMateControllerFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

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SetValues Method (IMateControllerFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMateControllerFeatureData Interface](#) : SetValues Method (IMateControllerFeatureData)

Sets the mate values for the specified mate position.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetValues( _  
    ByVal PositionName As System.String, _  
    ByVal Values As System.Object _  
)
```

Visual Basic (Usage)

```
Dim instance As IMateControllerFeatureData  
Dim PositionName As System.String  
Dim Values As System.Object  
  
instance.SetValues(PositionName, Values)
```

C#

```
void SetValues(  
    System.string PositionName,  
    System.object Values  
)
```

C++/CLI

```
void SetValues(  
& System.String^ PositionName,  
& System.Object^ Values  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PositionName

Name of mate position

Values

Array of values for all mates in PositionName

Visual Basic for Applications (VBA) Syntax

See [MateControllerFeatureData::SetValues](#).

Example

See the [IMateControllerFeatureData](#) examples.

Remarks

The number of items in Values must correspond to the number of mates passed to [IMateControllerFeatureData::Initialize](#) or the number of pre-selected mates.

See Also

[IMateControllerFeatureData Interface](#)
[IMateControllerFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31